(INCH-POUND)
A-A-58047
September 12, 1995

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
10.0 MHZ TO 26 5 GHZ

The General Services Administration has authorized
the use of this Commercial Item Description (CID) for all federal agencies

1. SCOPE

1 1 This CID describes a solid state general purpose 10 MHz to 26 5 GHz synthesized signal generator This CID is meant as a minimum requirement for the synthesized signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID

2. SALIENT CHARACTERISTICS

2 1 Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet the following description

2 1 1 The equipment shall be COTS equipment for bench-top, rack-mount and occasional portable use in widely diverse environmentally controlled locations

2 1 2 The enclosure or equipment case shall be an integral part of the equipment and provide protection from mechanical shock and conditions associated with bench-top use It shall have no openings on the top surface and will normally be used in an environmentally controlled area.

2 1 3 The color shall be that normally provided by the manufacturer

Beneficial comments recommendations additions, deletions, clarifications etc , and any data which may improve this document should be sent to Technology & Industrial Services Division, SA-ALC/TILDD, Bldg 171, Post C-12, 485 Quentin Roosevelt Rd , Kelly AFB, TX 78241-6425

AMSC N/A FSC 6625

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2 2 Safety and environmental The equipment shall meet all safety and environmental requirements as specified in MIL-T-28800 for the generator as stated herein

2 3 Electrical power sources The equipment shall operate from nominal commercial, military and shipboard power sources of 110/220 volts AC (+/- 10%) at line frequencies of 50 Hz to 60 Hz Power consumption shall not exceed 400 VA Transient state conditions shall meet MIL-T-28800 requirements for the generator specified herein

2 4 Temperature Requirements

2 4 1 Operating temperature The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to +50°C

2 4 2 Non-operating temperature The equipment shall meet its performance and accuracy requirements in a non-operating environment of -10°C to +60°C

2 5 Mean time between failures (MTBF) The equipment MTBF shall be 2,000 hours The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours

2 6 Pre-conditioning requirement Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase

2 7 Calibration and maintenance adjustments The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments These adjustments shall be provided by variable value components which are adjustable by the use of simple means The calibration by substitution of selected components or parts is unacceptable unless specifically approved The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours

2 8 Performance characteristics The synthesized signal generator shall be ASCII controllable language through an external IEEE 488 bus configuration per MIL-T-28800

2 8 1 Frequency The minimum frequency range of the synthesized signal generator shall be 10 MHz to 26 5 GHz

2 8 1 1 Frequency resolution

1 Hz,	0 01	to	7 0	GHz
2 Hz,	7 0	to	13 5	GHz
3 Hz,	13 5	to	20 0	GHz
4 Hz,	20 0	to	26 5	GHz

2 8 1 2 Accuracy The frequency accuracy shall be based on the internal reference oscillator aging rate which shall be 1×10^{-9} per day after 10 days of continuous operation

2 8 1 3 Switching speed Signal generator will switch frequencies in 50 milliseconds maximum

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2 8 1 4 Sweep time The minimum sweep time shall be at least 45 milliseconds.

2 8 2 Spectral purity

2 8 2 1 Harmonics The synthesized signal generator harmonics shall be -35 dBc or greater over the entire frequency range

2 8 2 2 Sub-harmonics The sub-harmonics shall be -25 dBc or greater to 18 GHz and -20 dBc or greater at 18 to 26 GHz

2 8 2.3 Spurious non-harmonics The spurious non-harmonics shall be at least -50 dBc over the entire frequency spectrum

2 8 2 4 Absolute SSB phase noise The absolute single sideband (SSB) phase noise at 26 GHz carrier at 100 KHz offset shall be at least -90 dBc per Hz

2 8 3 Output The minimum output level range into 50 ohm nominal shall be 8 dBm to -99 dBm for frequencies up to 20 GHz and 1 dBm to -99 dBm for frequencies from 20 to 26 5 GHz

2 8 3 1 Output resolution The synthesized signal generator output resolution shall be 0 1 dB

2 8 3 2 Output protection The synthesized signal generator shall have reverse power protection to −20 dBm

2 8 3 3 Output connector The RF output connector shall be APC-3 5 female

2 8 3 4 Output accuracy The output accuracy shall be +/- 2 dB

2 8 4 Sweep mode The signal generator shall provide analog and/or digital sweep capability

2 8 4 1 Center frequency/sweep width (ΔF)

2 8 4 1 1 Range 10 MHz to 26 5 GHz nominal (center frequency), 10 MHz to 26.5 GHz nominal (sweep width)

2 8 4 1 2 Resolution Approximately 0 1 % of sweep width (ΔF)

2 8 4 2 Start/stop frequency

2 8 4.2 1 Range: 10 MHz to 26 499 GHz nominal (start), 10 5 MHz to 26 5 GHz nominal (stop)

2 8 4 2 2 Resolution Same as center frequency/sweep width

2 8 4 3 Markers The signal generator shall have at least 5 independent markers

2 8 5 Pulse modulation The on/off ratio for the pulse modulation shall be 80 dB or greater

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2 8 5 1 Rise and fall time The rise and fall time between 10% and 90% of the pulse shall be 25 nanoseconds or less

2 8 5 2 Repetition rate The minimum repetition rate for a carrier frequency of 10 MHz to 26 5 GHz shall be 100 Hz to 1 MHz

2 8 5 3 Pulse width The minimum pulse width shall be at least 100 nanoseconds

2 8 6 Amplitude modulation

2 8 6 1 AM depth 90% typical Actual available depth will be greater than this in many cases and is determined by the difference of the maximum leveled power available at frequency of interest and -30 dBm

2 8 6 2 AM sensitivity (at a 1 KHz rate and 30% depth) 100%/V +5% AM depth is linearly controlled by varying input level between 0 and +1 V peak

2 8 6 3 AM bandwidth (30% depth PULSE off) 3 dB point 10 KHz

2 8 7 Frequency modulation

2 8 7 1 Modulation rates 50 KHz to 1 MHz

2 8 7 2 Peak deviation the lesser of 10 MHz or

Band 1	5 x Mod Rate
Band 2	10 x Mod Rate
Band 3	15 x Mod Rate
Band 4	20 x Mod Rate

2 8 7 3 Deviation accuracy +/- 10% at 100 KHz rate

2 8 8 Dimensions and weight The size and weight of the equipment shall be consistent with current commercial capabilities, reference to the following two paragraphs of this CID

2 8 8 1 Equipment dimensions The nominal dimensions for the synthesized signal generator should be approximately 8 inches high x 17 inches wide x 24 inches deep

2 8 8 2 Equipment weight The weight including all accessories (excluding manual) shall not exceed 80 lbs

3. REGULATORY REQUIREMENTS

3 1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23 403 of the Federal Acquisition Regulation (FAR)

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4. QUALITY ASSURANCE PROVISIONS

4 1 Certification The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description,and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices, and is the same product offered for sale in the commercial marketplace The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract

4 2 The equipment shall be bid sample tested within the accuracies limits and specifications herein

4 3 The contracting officer has the option of accepting or rejecting the product

5. PACKAGING

5 1 Preservation, packing, and marking shall be as specified in the contract or order

6. NOTES

6 1 Known acceptable products The following product is known to have previously met the requirements of this CID and need not resubmit bid samples or certification All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required

P/N	MANUFACTURER	NSN
PD91LDACID104 2	SA-ALC	6625-01-344-9277

6 2 Metric Products Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No 376, and all other requirements of this CID are met If a product is manufactured to metric dimensions exceeding the tolerances specified in the inch-pound units a request should be made to the contracting officer to determine if the product is acceptable

6 3 Federal and Military specifications are available from the Defense Documents Order Desk Building 4D 700 Robins Avenue Philadelphia PA 19111-5094

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MILITARY INTERESTS

Custodian
Air Force - 99

Preparing Activity
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CIVIL AGENCY COORDINATING ACTIVITY

GSA - FSS

Project No 6625-0801

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